

Closing the Handshake Gap: Toward a Scale-Invariant Port-Hamiltonian Safety Architecture, with Extreme-Case Instantiation in Medical Microrobotics

Durga Deepak Valluri*

Durga Prasad Sarilla†

Abstract

This paper proposes the Energetics-Aware Hamiltonian Control (EA-HC) framework, a vertically integrated architecture designed to resolve the “Handshake Gap”: the structural decoupling between high-level symbolic intent and low-level physical safety. While the individual components of energy-aware robotics are well-supported in existing literature, a unified stack bridging task-level semantics and physical thermodynamics remains an unaddressed gap. The proposed four-layer architecture translates symbolic task requirements into thermodynamic energy budgets, managed by a safe reinforcement learning layer and enforced through port-Hamiltonian (pH) energy tanks.

The primary aim is to investigate the plausibility of a scale-invariant control architecture in which deterministic passivity acts as a universal safety veto across diverse robotic embodiments. Medical microrobotics in stochastic, Brownian-dominated environments is selected as an extreme-case instantiation to examine whether the architecture remains well-defined and evaluable at the boundary of its operating range. The physical plant is modeled as a Stochastic Port-Hamiltonian System (SPHS), and the admissibility condition is extended by two stochastic energy buffers: an Itô correction and a Lévy jump correction, both derived directly from the Itô–Lévy formula. The scope of the thermodynamic veto is explicitly bounded: it enforces passivity against the robot’s own actuator outputs and internally generated stochastic disturbances. Geometric confinement against exogenous ambient forces (e.g., pulsatile blood-flow drift) is a distinct safety objective treated as future work.

This work proposes a foundation for further investigation into the EA-HC architecture as a scale-invariant framework unifying symbolic reasoning and physical safety across the full spectrum of robotic embodiments.

Note: Numerical simulation and hardware experiments are ongoing and will be reported in a subsequent version.

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*durgadeepak.valluri97@gmail.com

†dpsarilla@ksu.edu, Kansas State University

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1 Introduction

Modern robotics confronts a fundamental architectural tension. High-level task planners and reinforcement learning (RL) policies reason in the currency of symbolic rewards, abstract goals, and discrete state transitions [1, 2]. Physical actuators, by contrast, obey the laws of thermodynamics: energy must be conserved, dissipated, or stored, and passivity violations manifest as unstable oscillations, mechanical failure, or harm to human operators and biological tissue [3, 4].

We term this decoupling the *Handshake Gap*: the absence of a shared language that allows a high-level intent to be continuously translated into a physically valid actuation budget. The gap is not merely an implementation inconvenience; it is a structural safety hazard. A policy that hallucinates an optimal trajectory that is energetically infeasible will, in the absence of a physical veto, attempt to execute that trajectory through the actuators.

Existing approaches address fragments of this problem. Energy-aware control [5, 16] enforces passivity locally but does not connect to semantic task layers. Safe RL via Constrained Markov Decision Processes (CMDPs) [6, 7] learns safety boundaries statistically but cannot guarantee thermodynamic admissibility in closed form. Port-Hamiltonian systems [8, 9] provide a principled energy substrate but are not typically integrated with learning-based intent layers.

Scope clause. The EA-HC framework addresses the Handshake Gap by establishing energy as the universal safety currency between the symbolic and physical layers. The thermodynamic veto enforces passivity against the robot’s own actuator actions and internally generated stochastic disturbances. It does not, in its current form, guarantee spatial confinement against exogenous ambient forces (e.g., pulsatile blood flow, pressure gradients, or vessel-wall collisions). Geometric safety against such forces is a distinct objective and is identified as the primary direction for future work (Section 8).

Contributions. This paper makes the following contributions:

1. **Jump-diffusion SPHS extension.** We extend the stochastic port-Hamiltonian formulation of Cordoni et al. [11] – which covers general continuous semimartingales (excluding jump-discontinuous noise) – to the jump-diffusion regime by incorporating a compensated Poisson random measure following Applebaum [10]. Among the closely related SPHS works: Fang and Gao [13] study passivity-based stabilization under input disturbances for continuous-noise SPHS; Fang, Gao, and Dochain [66] extend this to stochastic weak passivity with nonvanishing noise; Cordoni et al. [59] combine Brownian SPHS with energy tanks for teleoperation; Cordoni et al. [63] develop discrete SPHS; and Cordoni et al. [62] characterize invariant measures for continuous SPHS. None of these works incorporate a compensated Poisson random measure in a finite-dimensional input-state-output port-Hamiltonian formulation. The present extension (Section 2) fills this gap following Applebaum’s Lévy calculus [10].
2. **Extended admissibility condition.** We extend the energy tank admissibility condition of Ferraguti et al. [4, 5] by two stochastic buffers ($\delta_{\text{Itô}}$ and $\delta_{\text{Lévy}}$) derived from the Itô–Lévy formula, enabling passivity guarantees in jump-diffusion environments. Zhang et al. [57] and Cordoni et al. [59] address related combinations of safe RL with energy tanks in the Brownian regime; the present paper adds the jump-diffusion buffer and the semantic intent layer.
3. **Unified four-layer architecture.** We integrate semantic intent translation, safe RL policy optimization, energy tank enforcement, and a jump-diffusion SPHS plant into a single closed

stack, with an extreme-case instantiation in medical microrobotics spanning seven orders of magnitude in operating scale.

The remainder of the paper is organized as follows. Section 2 reviews port-Hamiltonian systems and stochastic calculus prerequisites. Section 3 surveys related work. Section 4 presents the four-layer EA-HC architecture. Section 5 states and proves the main theoretical results. Section 6 instantiates the architecture for medical microrobotics. Section 7 discusses implications, limitations, and connections to broader research directions. Section 8 concludes and outlines future work.

2 Preliminaries

2.1 Port-Hamiltonian Systems

A finite-dimensional port-Hamiltonian (pH) system is defined by the state-space representation [8, 9]:

$$\dot{x} = [J(x) - R(x)]\nabla H(x) + g(x)u, \quad y = g(x)^T \nabla H(x), \quad (1)$$

where $x \in \mathbb{R}^n$ is the state, $H : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ is the Hamiltonian (total stored energy), $J(x) = -J(x)^T$ is the skew-symmetric interconnection matrix, $R(x) = R(x)^T \succeq 0$ is the positive semi-definite damping matrix, $g(x)$ is the input matrix, u is the control input, and y is the collocated output. The power balance identity follows immediately from (1):

$$\dot{H} = -\nabla H(x)^T R(x) \nabla H(x) + y^T u \leq y^T u, \quad (2)$$

confirming that the system is passive with supply rate $s(u, y) = y^T u$ [3].

2.2 Stochastic Port-Hamiltonian Systems

In stochastic operating environments, the deterministic plant (1) is generalized to a Stochastic Port-Hamiltonian System (SPHS) driven by both Brownian and Poisson noise. Cordoni et al. [11] established Itô formulations for SPHS driven by general continuous semimartingales (excluding jump processes); the jump-diffusion extension to include a compensated Poisson random measure has not appeared in the published port-Hamiltonian literature and is introduced here following Applebaum's [10] framework for Lévy-driven SDEs. The state evolves as:

$$dx = [J(x) - R(x)]\nabla H(x) dt + g(x)u dt + \sigma(x) dW_t + \int_{\mathcal{Z}} \gamma(x, z) \tilde{N}(dt, dz), \quad (3)$$

where W_t is a standard Wiener process, $\sigma(x)$ is the diffusion coefficient matrix, $\tilde{N}(dt, dz) = N(dt, dz) - \nu(dz) dt$ is the compensated Poisson random measure with Lévy intensity ν , $\mathcal{Z}$ is the mark space, and $\gamma(x, z)$ is the jump amplitude function.

Assumption 1 (Structural assumptions on noise and jumps). *Throughout this paper we impose the following structural conditions, which are physically natural for underdamped Langevin dynamics and magnetic microrobotics:*

- (S1) Separable Hamiltonian: $H(q, p) = K(p) + V(q)$ where $K(p) = \frac{1}{2m} \|p\|^2$ is the kinetic energy and $V(q)$ is a potential (not necessarily convex).
- (S2) Momentum-only noise: $\sigma(x) = \text{blockdiag}(0, \sigma_p(q))$, i.e., Brownian noise acts on momenta only.

(S3) Momentum-only jumps: $\gamma(x, z) = (0, \gamma_p(z))$ with γ_p independent of (q, p) , i.e., jump events displace momenta only.

(S4) Finite second moments: $\int_{\mathcal{Z}} \|\gamma_p(z)\|^2 \nu(dz) < \infty$.

Under (S1)–(S3), the jump map $(q, p) \mapsto (q, p + \gamma_p(z))$ has unit Jacobian and therefore preserves the symplectic form $dq \wedge dp$. Under (S4), the compensated Poisson integral is a square-integrable martingale [10].

2.3 Itô–Lévy Formula

For a twice continuously differentiable function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ and a process satisfying (3) (driven by the compensated measure $\tilde{N}$), the Itô–Lévy formula in compensated form gives [10, 12]:

$$\begin{aligned} dV(x) = & \nabla V(x)^T dx^c + \frac{1}{2} \text{tr}[\sigma(x)^T \nabla^2 V(x) \sigma(x)] dt \\ & + \int_{\mathcal{Z}} [V(x + \gamma(x, z)) - V(x) - \nabla V(x)^T \gamma(x, z)] \nu(dz) dt \\ & + \int_{\mathcal{Z}} [V(x + \gamma(x, z)) - V(x)] \tilde{N}(dt, dz), \end{aligned} \quad (4)$$

where $dx^c = [(J(x) - R(x))\nabla H(x) + g(x)u]dt + \sigma(x)dW_t$ denotes the continuous semimartingale part of dx . The last line is a zero-mean martingale increment under the square-integrability condition $\int_{\mathcal{Z}} \mathbb{E}[(V(x + \gamma) - V(x))^2] \nu(dz) < \infty$ [10]. Applying (4) with $V = H$ yields the stochastic energy balance used in Section 4.5 and Proposition 1.

2.4 Energy Tanks

Energy tanks [4, 5, 39] augment a pH system with a scalar energy reservoir T that mediates the storage and release of interaction energy. The tank dynamics are:

$$\dot{T} = -u_T^T \dot{x}_T, \quad T \in [T_{\min}, T_{\max}], \quad (5)$$

where u_T and $\dot{x}_T$ are the tank port variables. The tank enforces a hard energy ceiling: if the remaining tank level would fall below $T_{\min}$ after executing a proposed control action, that action is vetoed and replaced by a safe fallback [4].

3 Related Work

3.1 Port-Hamiltonian Control and Passivity

The port-Hamiltonian framework was formalized by van der Schaft and Maschke [8, 9] as a coordinate-free, energy-centric representation of physical systems. Passivity-based control (PBC) leverages the pH structure to design controllers that shape the closed-loop energy landscape [3, 14]. Energy tanks, introduced by Ferraguti et al. [4, 5], extended this idea by decoupling the energy budget from the control design, enabling variable-impedance telemanipulation while preserving passivity.

3.2 Safe Reinforcement Learning

The CMDP framework [7] extends Markov Decision Processes with inequality constraints on cumulative costs, providing the canonical formulation for Safe RL [6]. Recent work has integrated

energy constraints into RL pipelines [15]. Most closely related to the present work, Zhang et al. [57] combine energy-tank passivity filtering with CMDP-based safe RL for contact-rich manipulation in a deterministic macro-scale setting (preprint, under review); Zanella et al. [58] learn passive policies using virtual energy tanks via RL; Piccinelli et al. [67] explicitly couple an energy tank with a CMDP-formulated RL critic and LQR convergence layer for cyber-physical systems. None of these works include a stochastic port-Hamiltonian plant with jump-diffusion noise or a semantic intent layer translating task specifications into energy budgets. Cordini, Di Persio, and Muradore [59] combine Brownian SPHS with energy tanks, providing a foundational result that the present paper extends to the jump-diffusion regime. For the semantic-to-safety-constraint pattern, Brunke et al. [68] provide the closest analogue using CBFs rather than energy ceilings; the present Layer 1 mapping to a thermodynamic energy budget is the specific composition not found in prior work. Reward shaping via physical priors [2] offers complementary approaches but does not yield deterministic safety certificates.

3.3 Stochastic Port-Hamiltonian Systems

Stochastic extensions of pH systems have been studied by Cordini et al. [11], who established Itô formulations for SPHS driven by general continuous semimartingales (subsuming Brownian motion but excluding jump-discontinuous Lévy processes); recent work from the same group [62] shows that the Fokker–Planck equation of an SPHS defines an infinite-dimensional deterministic PHS on L^2_ρ and derives explicit invariant measures. Fang and Gao [13] analyzed stabilization of input-disturbed SPHS via passivity; Fang, Gao, and Dochain [66] extend this to stochastic weak passivity with nonvanishing noise. Cordini et al. [59] combine energy tanks with SPHS (Brownian regime), and Cordini et al. [63] develop discrete stochastic PHS. The key distinction among these works is as follows. Cordini et al. [11] treat the most general continuous-semimartingale noise but explicitly exclude jump-discontinuous processes. Fang and Gao [13] and Fang et al. [66] study passivity-based stabilization but do not extend to the jump-diffusion regime. Cordini et al. [59] combine SPHS with energy tanks but in the Brownian-only regime. Cordini et al. [63] work in discrete time without the compensated Poisson random measure. None of these works therefore constitute the finite-dimensional input-state-output jump-diffusion SPHS formulation introduced in Section 2, which follows Applebaum’s Lévy calculus [10].

3.4 Medical Microrobotics

Untethered microrobots operating in physiological environments face stochastic disturbances several orders of magnitude larger relative to their body scale than macro-scale robots [23, 24]. Magnetic actuation schemes [23] and acoustic propulsion [25] have been demonstrated. Passivity-based safety for magnetic microrobots has been studied via Time-Domain Passivity Control (TDPC); Duygu et al. [69] demonstrate real-time TDPC-based teleoperation of magnetic microrobots over intercontinental distances. However, the port-Hamiltonian energy-tank formulation – which provides a unified thermodynamic veto layer that integrates with semantic task specifications and safe RL – has not been instantiated for medical microrobotics, and the extension to jump-diffusion SPHS environments remains an open problem that the EA-HC framework addresses.

3.5 Control Barrier Functions and Geometric Safety

Control Barrier Functions (CBFs) [17] provide a complementary safety paradigm: rather than bounding energy, they enforce forward invariance of safe sets in the state space, providing geometric confinement guarantees. CBF-based quadratic programs have been applied to obstacle avoidance

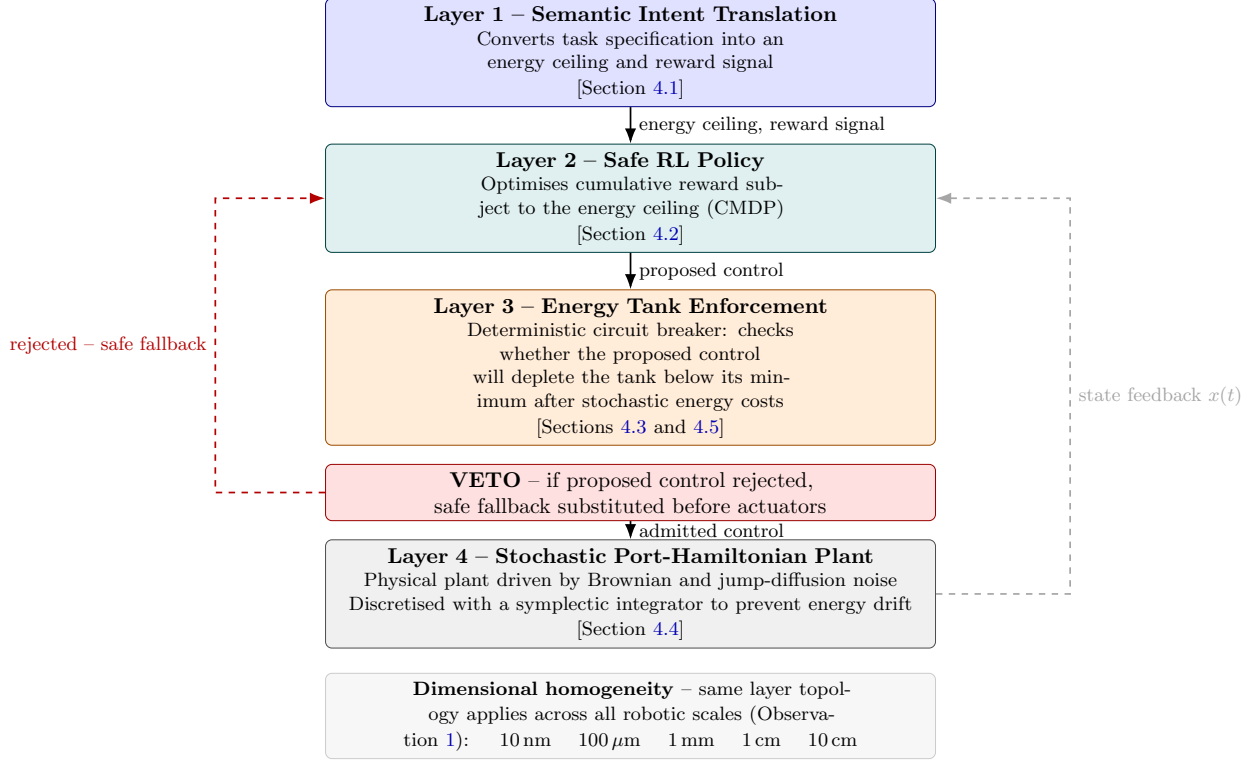


Figure 1: The EA-HC four-layer architecture. Downward arrows carry the signal flow from symbolic intent to physical actuation. The red dashed veto path returns a safe fallback u_{safe} to Layer 2 whenever the proposed control u_{π} falls outside the admissible set $\mathcal{U}_{\text{EAHC}}$. The grey dashed arrow carries state feedback $x(t)$ from Layer 4 back to Layers 2 and 3. The admissibility condition (12) is dimensionally homogeneous in Joules, enabling the same layer topology to be instantiated across seven orders of magnitude in operating scale (Observation 1).

[17] and multi-agent safety [18]. The EA-HC framework and CBF approaches address orthogonal objectives: thermodynamic passivity (power-balance enforcement via the energy veto) versus spatial confinement (forward invariance of safe sets). Their integration is identified as the primary direction for future work.

4 The EA-HC Architecture

The EA-HC framework is organized as a four-layer vertical stack. Each layer communicates with adjacent layers through precisely defined interface signals, described in Sections 4.1–4.4. Figure 1 provides an overview of the complete architecture, including signal flow, the veto path, and the stochastic buffer equations.

4.1 Layer 1: Semantic Intent Translation

The topmost layer is the entry point for human intent into the architecture. It receives a task specification – expressed in natural language, a formal task algebra, or a structured symbolic representation – and translates it into two signals that the layers below can act on: a scalar energy

ceiling $E_{\max} \in \mathbb{R}_{>0}$ and a task reward signal r . The mapping:

$$\mathcal{M} : \text{TaskSpec} \rightarrow (E_{\max}, r) \quad (6)$$

performs a function that has no direct counterpart in standard robot control pipelines. In most architectures, task specifications influence only the reward or cost function that a planner optimises; they do not constrain the energy budget of the actuators [1, 40]. The mapping $\mathcal{M}$ closes this gap by treating task semantics as a source of thermodynamic information. A task labelled as high-contact (e.g., manipulation against a stiff environment) receives a higher $E_{\max}$ than a task labelled as proximity-only (e.g., inspection without tissue contact) [5, 42]. The physical safety envelope is therefore not fixed at design time but is adapted continuously to the semantic content of the current task.

The novelty of Layer 1 lies in the *concept* of treating task-semantic information as a source of thermodynamic budget constraints – a composition not present in existing architectures. In the current implementation, $\mathcal{M}$ is instantiated as a rule-based lookup informed by the constraint-based task specification framework of Borghesan et al. [19] and the iTaSC formalism of Vanthienen et al. [20]. These frameworks provide a structured vocabulary for expressing task constraints that can be systematically associated with energy bounds. The rule-based approach is deliberately conservative: it is verifiable by inspection and introduces no learned uncertainty into the safety-critical energy ceiling. The principal limitation – and the motivation for the LLM-based extension described in Section 7.5 – is that the mapping is domain-specific and requires manual authoring for each new task class. This is acknowledged explicitly in Section 7.7.6.

4.2 Layer 2: Safe RL Policy

Layer 2 is responsible for generating task-level behaviour within the energy budget supplied by Layer 1. A Constrained Markov Decision Process (CMDP) policy π [7] operates on the current system state x and the energy ceiling $E_{\max}$, optimising cumulative reward subject to a long-run energy constraint:

$$\max_{\pi} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right] \quad \text{s.t.} \quad \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t H(x_t) \right] \leq E_{\max}, \quad (7)$$

where $\gamma \in (0, 1)$ is the discount factor shared with the reward objective, ensuring the constraint sum converges and remains dimensionally consistent with $E_{\max}$ in Joules. The discount factor is necessary: without it, $\sum H(x_t)$ has units of Joules $\times$ steps rather than Joules, and the constraint is ill-posed. The energy ceiling $E_{\max} \in \mathbb{R}_{>0}$ produced by Layer 1 enters (7) directly as the right-hand side bound, establishing a consistent energy language from the semantic layer through to the RL policy.

The CMDP formulation was chosen over unconstrained RL because it provides a principled mechanism for encoding the energy ceiling as a hard constraint on the policy’s behaviour over time, rather than as a soft penalty term in the reward that a sufficiently capable policy might learn to trade away [6, 41]. The constraint is expressed in terms of the Hamiltonian $H(x_t)$, which is the same quantity tracked by the energy tank in Layer 3, creating a consistent energy language across both layers.

The policy outputs a nominal control u_{π} at each time step, which is forwarded to Layer 3 for admissibility screening before it reaches the actuators. It is important to note that Layer 2 provides statistical safety in the sense of expected constraint satisfaction over a trajectory [6], but it does not provide a per-step guarantee. A policy that satisfies (7) in expectation may still propose controls that are instantaneously inadmissible from the tank’s perspective. This failure mode is well-documented

in the safe RL literature under distributional shift, where the policy encounters states outside its training distribution and constraint satisfaction is no longer guaranteed [6, 41]. Layer 3 exists precisely to intercept such proposals. The state $x(t)$ measured from the plant feeds back into the policy at every step, closing the loop between physical reality and the learned behaviour.

Remark 1 (Consistency between $E_{\max}$ and $T_{\max}$). *The energy ceiling $E_{\max}$ produced by Layer 1 enters the discounted CMDP constraint (7) as a present-value occupancy-weighted Joule budget. If the Hamiltonian is pathwise bounded by the tank ceiling, $H(x_t) \leq T_{\max}$ for all t , then $\mathbb{E}[\sum_{t=0}^{\infty} \gamma^t H(x_t)] \leq T_{\max}/(1 - \gamma)$. For the CMDP constraint to be satisfiable, a sufficient consistency condition is:*

$$T_{\max} \leq \frac{E_{\max}}{1 - \gamma}. \quad (8)$$

Equivalently, $(1 - \gamma)T_{\max} \leq E_{\max}$: the per-step marginal energy cost discounted to present value must not exceed the semantic budget. This condition constrains the system engineer’s choice of $T_{\max}$ given the semantically derived $E_{\max}$ and the RL discount factor γ ; it should be verified at design time before training the CMDP policy. The minimum threshold $T_{\min} \in (0, T_{\max})$ provides a safety margin absorbing estimation errors in $\delta_{\text{Itô}}$ and $\delta_{\text{Lévy}}$.

Remark 2 (Relationship between the CMDP constraint and the tank condition). *The CMDP constraint (7) bounds the cumulative sum of $H(x_t)$ in expectation – a soft, time-averaged objective that the RL policy learns to respect through training. The tank admissibility condition (12) is a per-step, hard constraint on the energy flow out of the tank. The two are not equivalent and do not imply one another: a policy satisfying (7) may still propose individual controls u_{π} that violate (12) on particular time steps, and vice versa. Their roles are complementary: the CMDP reduces the statistical frequency with which u_{π} falls outside $\mathcal{U}_{\text{EAHC}}$, while Layer 3 ensures that any such violation is intercepted before reaching the actuators. The formal composition of these two safety mechanisms – in particular, whether CMDP feasibility reduces the worst-case veto rate – is left as an open problem.*

4.3 Layer 3: Energy Tank Enforcement

Layer 3 is the deterministic safety core of the EA-HC architecture. It implements the energy tank originally introduced by Ferraguti et al. [4, 5, 39] and extends it with two stochastic energy buffers derived from the Itô–Lévy formula (Section 4.5). The layer maintains a scalar energy reservoir T_t that tracks how much energy remains available for expenditure. At every control cycle, before any control signal reaches the actuators, Layer 3 evaluates whether the proposed control u_{π} from Layer 2 is admissible: that is, whether executing it would leave the tank above the minimum threshold $T_{\min}$ after accounting for the power drawn by the control action and the stochastic energy taxes imposed by the operating environment.

The architectural significance of this layer is that it provides a safety guarantee that is independent of the quality or correctness of the upstream policy. Even if the CMDP policy in Layer 2 produces a control that is locally optimal according to its learned value function but globally inadmissible from an energy standpoint – whether due to incomplete training, distributional shift, or simply the inherent imprecision of statistical learning [6, 41] – Layer 3 intercepts that control and substitutes the minimum-norm passive fallback before any physical harm can occur. This separation between intent generation (Layer 2) and intent enforcement (Layer 3) parallels the principle of runtime monitoring in safety-critical systems, where a verified supervisor intercepts unsafe commands from an unverified controller [43, 44].

The tank state T_t is updated at every time step based on the actual energy consumed by the admitted control and the stochastic energy injections estimated from the current state. It is bounded below by $T_{\min} > 0$, which is a design parameter set by the system engineer to provide a safety margin against estimation errors in the stochastic buffers. When the tank approaches depletion under sustained disturbance, the veto rate increases, which serves as a system-level signal that the environment is more energetically demanding than the current policy anticipated [5]. The extended admissibility condition is derived in detail in Section 4.5.

4.4 Layer 4: Stochastic Port-Hamiltonian Plant

Layer 4 is the physical plant: the robotic system whose dynamics the architecture is designed to control safely. Within the EA-HC framework, the plant is modelled as an SPHS (3), which extends the deterministic pH formulation of Section 2 with both continuous Brownian diffusion and discrete jump events. This modelling choice is a deliberate commitment about which physical phenomena the safety architecture takes responsibility for. The Brownian term $\sigma(x) dW_t$ captures diffusive disturbances such as thermal noise, sensor jitter, and unmodelled friction that are continuous in time [45, 46]. The jump term $\int_{\mathcal{Z}} \gamma(x, z) \tilde{N}(dt, dz)$ captures episodic impulsive events such as collisions, sudden load changes, or, in the microrobotics context, pressure impulses from pulsatile blood flow [10, 27].

Both noise sources enter the system through the port structure of the pH model – they are intrinsic to the plant’s operating environment rather than exogenous inputs from outside the system boundary [8, 11]. This is the precise sense in which the scope of the thermodynamic veto is bounded: Layer 3 accounts for the energy injected by these port-connected disturbances, but it does not account for forces that act on the robot without passing through the pH port (e.g., a river current pushing a passive body). The distinction is discussed in detail in the limitations (Section 7.7.1).

The admitted control u_{adm} from Layer 3 drives the plant through the input matrix $g(x)$. The resulting state $x(t)$ is measured and fed back to both Layer 2 (where it updates the policy’s state estimate) and Layer 3 (where it updates the tank’s energy accounting and the stochastic buffer estimates for the next cycle). This feedback closes the loop and allows the architecture to adapt continuously to the evolving energy landscape of the physical system. The digital implementation of this feedback loop uses the symplectic Euler–Maruyama scheme described in Section 4.6, which preserves the Hamiltonian structure of the continuous-time model and prevents the artificial energy drift that would otherwise corrupt the tank’s energy ledger.

4.5 Extended Admissibility Condition

The admissibility condition screens whether the proposed control u_π is safe to admit. Applying the Itô–Lévy formula (4) to the Hamiltonian H along trajectories of (3), the expected energy injected by stochastic disturbances over one control interval Δt decomposes into two buffers.

Itô buffer. The continuous Brownian contribution, following the infinitesimal generator analysis of Cordoni et al. [11]:

$$\delta_{\text{Itô}} = \frac{1}{2} \text{tr}[\sigma(x)^T \nabla^2 H(x) \sigma(x)] \cdot \Delta t. \quad (9)$$

Lévy buffer. Under Assumption 1 (S1)–(S3), jump events displace momenta only: $\gamma = (0, \gamma_p(z))$. The second-order Taylor remainder of H with respect to a momentum-only displacement reduces

exactly (with no approximation, since K is quadratic in p):

$$H(x + \gamma) - H(x) - \nabla H^T \gamma = K(p + \gamma_p) - K(p) - \nabla_p K^T \gamma_p = \frac{\|\gamma_p(z)\|^2}{2m} \geq 0. \quad (10)$$

The non-negativity holds unconditionally – no convexity of $V(q)$ is required – because $V(q)$ drops out entirely from the p -Taylor expansion. Integrating against ν over one control interval Δt :

$$\delta_{\text{Lévy}} = \Delta t \int_{\mathcal{Z}} \frac{\|\gamma_p(z)\|^2}{2m} \nu(dz) \geq 0. \quad (11)$$

The finite-second-moment condition (S4) ensures $\delta_{\text{Lévy}} < \infty$. The result in (10)–(11) is exact under Assumption 1; no small-jump approximation is required.

Admissibility condition. The total energy extracted from the tank over one control interval Δt has two components: the deterministic power drawn by the admitted control, $y^T u_\pi \Delta t = \nabla H(x)^T g(x) u_\pi \Delta t$, and the stochastic energy injected by the noise, $\delta_{\text{Itô}} + \delta_{\text{Lévy}}$. A control u_π is admitted if and only if the tank level T remains above the minimum threshold after both deductions:

$$\mathcal{U}_{\text{EAHC}} = \left\{ u_\pi \mid T - \nabla H(x)^T g(x) u_\pi \Delta t - (\delta_{\text{Itô}} + \delta_{\text{Lévy}}) \geq T_{\min} \right\}. \quad (12)$$

The set $\mathcal{U}_{\text{EAHC}}$ is a half-space in u_π (for fixed x and T), so the minimum-norm element satisfying (12) exists and is unique whenever the set is non-empty [4]. If $u_\pi \notin \mathcal{U}_{\text{EAHC}}$, Layer 3 substitutes the minimum-norm passive fallback $u_{\text{safe}} = \arg \min \|u\|$ subject to (12), which reduces to $u_{\text{safe}} = 0$ when $T - \delta_{\text{Itô}} - \delta_{\text{Lévy}} < T_{\min}$ regardless of u_π – i.e., when the stochastic debits alone would deplete the tank, no control action is admissible and the robot is immobilised until the tank recharges. When $\nabla H(x)^T g(x) = 0$ (i.e., at energy extrema such as equilibria of regulation tasks), the power-flow term vanishes and the admissibility condition reduces to the pure stochastic constraint $T - \delta_{\text{Itô}} - \delta_{\text{Lévy}} \geq T_{\min}$, independent of u_π ; in this case any control is equally (in)admissible and the fallback is $u_{\text{safe}} = 0$. The stochastic buffers and the power-flow term together account for all energy leaving the tank; no exogenous non-port terms are included, in keeping with the bounded scope of the thermodynamic veto.

4.6 Symplectic Euler–Maruyama Discretization

For digital implementation, the SPHS (3) is discretized using the symplectic Euler–Maruyama scheme [21, 22]:

$$\begin{aligned} p_{k+1} &= p_k - \nabla_q H(q_k, p_{k+1}) \Delta t + g_p(x_k) u_k \Delta t + \sigma_p(x_k) \Delta W_k, \\ q_{k+1} &= q_k + \nabla_p H(q_k, p_{k+1}) \Delta t, \end{aligned} \quad (13)$$

where (q, p) are generalized coordinates and momenta, and $\Delta W_k \sim \mathcal{N}(0, \Delta t)$.

Implicit step resolution. The momentum update in (13) is implicit in p_{k+1} through $\nabla_q H(q_k, p_{k+1})$. For the separable Hamiltonian $H = K(p) + V(q)$ of Assumption 1(S1), $\nabla_q H = \nabla_q V(q_k)$ is independent of p_{k+1} , so the implicit step reduces to an *explicit* update:

$$p_{k+1} = p_k - \nabla_q V(q_k) \Delta t + g_p(x_k) u_k \Delta t + \sigma_p(x_k) \Delta W_k. \quad (14)$$

No Newton iteration or fixed-point solve is required for any instantiation satisfying Assumption 1(S1). For non-separable Hamiltonians, the implicit step must be resolved by a fixed-point iteration or linearization; this should be verified for each new instantiation.

Structural validity. Stochastic symplecticity in the mean-square sense holds for separable Hamiltonians $H = K(p) + V(q)$ with additive noise on p only [60] – the setting of the microrobotics instantiation (Section 6). For state-dependent (multiplicative) σ , fully implicit schemes are required [61]. For the jump term, symplecticity requires the jump map $x \mapsto x + \gamma(x, z)$ to be symplectic; this holds for momentum-only jumps $\gamma = (0, \gamma_p)$ under Assumption 1(S3) (Jacobian $= I$, so $dq \wedge dp$ is preserved), but not generically. These assumptions are satisfied in the microrobotics case study and should be verified for other instantiations.

The symplectic structure preserves the Poincaré invariants of the Hamiltonian flow up to discretization error of order $\mathcal{O}(\Delta t^2)$ [22], preventing artificial energy drift that would contaminate the tank’s energy accounting.

5 Theoretical Analysis

5.1 Compositional Passivity

Proposition 1 (Compositional Passivity). *Consider the EA-HC stack with plant (3), tank dynamics (5), and admissibility condition (12). If at every time step the admitted control satisfies $u_{\text{adm}} \in \mathcal{U}_{\text{EAHC}}$, then the closed-loop system is passive with supply rate $s(u, y) = y^T u$ in expectation:*

$$\mathbb{E}[H(x(t))] \leq H(x(0)) + \int_0^t \mathbb{E}[y(s)^T u_{\text{adm}}(s)] ds. \quad (15)$$

Proof. Step 1: Non-explosion and localization. Under Assumption 1 and global linear growth of the coefficients $[J - R, g, \sigma_p, \gamma_p]$, the SDE (3) admits a unique global càdlàg strong solution with $\tau_\infty = \infty$ a.s. [10, 65]. Define stopping times $\tau_n = \inf\{t \geq 0 : H(x(t)) \geq n\}$ for $n \in \mathbb{N}$. Since $H(x_0) \leq E_{\max} < \infty$ and the solution is non-explosive, $\tau_n \nearrow \infty$ a.s. by Applebaum [10] (Theorem 6.2.3). On the stopped interval $[0, \tau_n]$, the Hamiltonian is bounded by n , so the square-integrability condition $\int_{\mathcal{Z}} \mathbb{E}[(H(x^{\tau_n} + \gamma) - H(x^{\tau_n}))^2] \nu(dz) < \infty$ holds under (S4), and the compensated Poisson integral is a square-integrable martingale (Applebaum [10], Proposition 4.4.1). The stochastic passivity in expectation framework of Lanchares and Haddad [64] establishes the equivalence between the energy-function inequality and the supermartingale property for Itô diffusions; we extend this to the jump-diffusion case via the Itô–Lévy formula following the same argument.

Step 2: Per-step discrete inequality. At each Euler–Maruyama step k on the stopped trajectory, applying the Itô–Lévy formula (4) to H and using Assumption 1(S1)–(S3):

$$\mathbb{E}[H(x_{k+1}^{\tau_n})] - \mathbb{E}[H(x_k^{\tau_n})] \leq \Delta t \mathbb{E}[\nabla H^T g u_{\text{adm},k}] + \mathbb{E}[\delta_{\text{Itô}}] + \mathbb{E}[\delta_{\text{Lévy}}], \quad (16)$$

where the Brownian increment $\nabla H^T \sigma_p \Delta W_k$ and the compensated Poisson increment $\int [H(x + \gamma) - H(x)] \tilde{N}(\Delta t, dz)$ have zero expectation by the martingale property of Step 1. The admissibility condition (12) ensures the tank absorbs $\delta_{\text{Itô}} + \delta_{\text{Lévy}}$ per step before releasing net power $\nabla H^T g u_{\text{adm},k} \Delta t$; formally, the tank balance at step k gives $\nabla H^T g u_{\text{adm},k} \Delta t + \delta_{\text{Itô}} + \delta_{\text{Lévy}} \leq T_k - T_{\min}$, so the right-hand side of (16) is bounded by the external supply $\Delta t \mathbb{E}[y_k^T u_{\text{adm},k}]$.

Step 3: Summation and limit. Summing (16) over $k = 0, \dots, \lfloor t/\Delta t \rfloor - 1$:

$$\mathbb{E}[H(x^{\tau_n}(t))] \leq H(x_0) + \sum_k \Delta t \mathbb{E}[y_k^T u_{\text{adm},k}].$$

As $\Delta t \rightarrow 0$ the Riemann sum converges to $\int_0^t \mathbb{E}[y(s)^T u_{\text{adm}}(s)] ds$ under standard integrability of u_{adm} . Taking $n \rightarrow \infty$ and applying the monotone convergence theorem (since $H \geq 0$ and $\tau_n \nearrow \infty$ a.s.) yields (15). $\square$

5.2 Stochastic Safety Guarantee

Proposition 2 (Stochastic Energy Bound). *Under the conditions of Proposition 1 – specifically: (i) the admissibility condition (12) is enforced at every discrete time step; (ii) Assumption 1 holds; (iii) the plant model $H(x)$ is faithful to the true system; and (iv) the CMDP policy π has converged to satisfy constraint (7) – if $T_0 \geq T_{\min}$ and $H(x_0) \leq E_{\max}$, and the admitted control satisfies $\|u_{\text{adm}}(t)\| \leq u_{\max} < \infty$ for all $t \geq 0$, then for all $t \in [0, T]$:*

$$\Pr[H(x(t)) > E_{\max}] \leq \frac{H(x_0) + u_{\max} \cdot \|g\|_{\infty} \cdot \|\nabla H\|_{\infty} \cdot T}{E_{\max}}. \quad (17)$$

Proof. The boundedness assumption $\|u_{\text{adm}}\| \leq u_{\max}$ ensures that the supply-rate integral satisfies $\int_0^T \mathbb{E}[y(s)^T u_{\text{adm}}(s)] ds \leq u_{\max} \cdot \|g\|_{\infty} \cdot \|\nabla H\|_{\infty} \cdot T < \infty$. Apply Markov’s inequality [47] to (15) with $V = H \geq 0$ to obtain (17). The bound is finite-horizon and conservative: it ignores the variance of H and the tail structure of the Lévy noise. Tighter bounds via the Lévy exponent (Cramér–Chernoff) are available [10] and are identified as a direction for future work. $\square$

5.3 Dimensional Homogeneity and Scale-Agnostic Formulation

Observation 1 (Dimensional Homogeneity). *The admissibility condition (12) is dimensionally homogeneous in energy units (Joules) and is independent of the absolute magnitude of the system coordinates. Consequently, the same EA-HC layer topology applies to robotic systems across arbitrary operating scales, provided the Hamiltonian H and diffusion coefficients σ are specified consistently with the system’s energy domain [3, 48].*

Proof. The tank level T , the buffers $\delta_{\text{Itô}}$ and $\delta_{\text{Lévy}}$, and the minimum threshold $T_{\min}$ are all measured in Joules. The admissibility condition (12) is a scalar inequality in energy, with no explicit dependence on length, mass, or time scales [3]. The same inequality therefore applies verbatim at any scale; only the numerical values of H , σ , and ν change. This is a dimensional consistency observation: a formal invariance theorem under a specified nondimensionalization group would require additional assumptions on the scaling of H , σ , and ν across the macro-to-micro range and is left as future work. $\square$

6 Medical Microrobotics: Extreme-Case Instantiation

Medical microrobotics represents an extreme-case instantiation of the EA-HC framework: operating scales span from centimeters (catheter-tipped robots [23]) to nanometers (DNA-origami walkers [28]), thermal noise dominates actuation forces below roughly $10 \mu\text{m}$ [29], and the operating environment (blood, interstitial fluid, cerebrospinal fluid) imposes both stochastic Brownian disturbances and episodic jump-like perturbations from pulsatile flow.

6.1 SPHS Instantiation for Microswimmers

A magnetic microswimmer is modeled as a rigid body with generalized coordinates $q = (p_x, p_y, p_z, \varphi)^T$ (position and orientation) and momenta $p = m\dot{q}$. The Hamiltonian is:

$$H(q, p) = \frac{1}{2m} \|p\|^2 + V_{\text{mag}}(q), \quad (18)$$

Table 1: Scale-agnostic instantiation of the EA-HC architecture. [†]At 10 nm quantum effects become non-negligible and the classical SPHS Hamiltonian assumption may break; this row is conjectural and included for illustrative completeness only.

System	Scale	H domain	σ origin	Jump source
Surgical robot	10 cm	Rigid-body KE	Encoder noise	Tool contact
Soft exoskeleton	1 cm	Elastic PE	Gait variability	Stumble events
Catheter robot	1 mm	Bending energy	Blood drag	Pulsatile flow
Microswimmer	100 μm	Thermal KE	FDT (Eq. 19)	Pulse impulse
DNA nanobot [†]	10 nm	Binding energy	Thermal agitation	Molecular collision

where V_{mag} is the magnetic potential energy from the external actuation coils [23]. The diffusion coefficient is set by the fluctuation-dissipation theorem [26]:

$$\sigma = \sqrt{2k_B T_{\text{env}} \zeta} I, \quad (19)$$

where k_B is Boltzmann’s constant, T_{env} is the fluid temperature, and ζ is the drag coefficient from Stokes’ law. Jump events $\gamma(x, z)$ model pulsatile pressure impulses with Lévy intensity ν calibrated from physiological data [27]. Assumption 1(S1)–(S4) is satisfied: the Hamiltonian (18) is separable, the diffusion (19) acts on momenta only, pulsatile impulses displace momenta only, and the finite-second-moment condition holds for physiologically calibrated ν [27].

6.2 Buffer Evaluation at the Microscale

Substituting (18) into (9) with the diffusion coefficient (19) and using $\nabla^2 H = \frac{1}{m} I$ (from the quadratic kinetic energy term) yields the Itô buffer directly:

$$\delta_{\text{Itô}} = \frac{k_B T_{\text{env}} \zeta}{m} \cdot n \cdot \Delta t, \quad (20)$$

where n is the state dimension.

For the Lévy buffer, under Assumption 1(S1)–(S3), pulsatile pressure impulses displace the swimmer’s momentum only: $\gamma(x, z) = (0, \gamma_p(z))$ with $\gamma_p(z) = z$. By (10) the second-order remainder is exactly $\|z\|^2/(2m)$, so:

$$\delta_{\text{Lévy}} = \Delta t \int_0^\infty \frac{\|z\|^2}{2m} \nu(dz). \quad (21)$$

This result is exact under Assumption 1 – no approximation is invoked.

Both $\delta_{\text{Itô}}$ and $\delta_{\text{Lévy}}$ are computable from known physical constants and calibrated physiological parameters [27], confirming that the admissibility condition (12) is evaluable at the microscale. The tightness of these bounds – i.e., whether they are conservative or near-exact in practice – will be assessed in forthcoming simulation studies.

6.3 Scale Comparison

Table 1 summarizes the EA-HC instantiation across five robotic embodiments spanning seven orders of magnitude in characteristic length.

7 Discussion

7.1 Architectural Significance

The EA-HC framework is, to our knowledge, the first architecture to integrate all four layers (semantic intent, safe RL, energy tank enforcement, and stochastic pH plant) into a single closed stack. Each layer has antecedents in the literature [4, 7, 11, 19]; the contribution is the vertical integration and the explicit interface definitions between layers.

The Handshake Gap is not a gap in mathematics but a gap in architecture. Energy-aware controllers exist [5, 16]; safe RL methods exist [6, 41]; stochastic pH models exist [11, 13]. What has been missing is a load-bearing interface between them that allows a symbolic task specification to flow downward as a thermodynamic budget and a passivity certificate to flow upward as a safety guarantee. The mapping $\mathcal{M}$ (Layer 1 to Layer 2) and the admissibility condition (Layer 2 to Layer 3) are the two novel interfaces that close this gap.

7.2 Structural vs. Statistical Safety Enforcement

Unlike traditional Safe RL frameworks that rely on a CMDP to learn safety boundaries statistically, the EA-HC architecture enforces safety structurally. The Handshake Gap is fundamentally a language barrier: high-level agents operate in symbolic rewards, while physical systems operate in Joules. By establishing energy as the universal currency, the EA-HC provides a deterministic circuit breaker. Even if an RL policy produces a nominally optimal trajectory that is physically destructive, the Layer 3 energy tank acts as a final arbiter, ensuring that passivity cannot be violated by the admitted control.

This distinction matters for certification. Statistical safety (CMDP) provides bounds on expected constraint violation but cannot rule out individual catastrophic episodes [6]. Structural safety (the energy veto) provides a hard guarantee per control cycle, conditional on model fidelity (see Section 7.7.2), and is therefore closer in spirit to the deterministic safety certificates sought by standards such as IEC 61508 and ISO 10218 for safety-critical robotic systems [44]. The two mechanisms are complementary: the CMDP reduces the frequency with which Layer 3 must veto, while Layer 3 ensures that any veto failure at the CMDP level does not propagate to the actuators.

7.3 Applications Across Robotic Domains

The dimensional homogeneity of the admissibility condition (Observation 1) implies that the EA-HC architecture is deployable, with appropriate Hamiltonian and diffusion parameterization, across a wide range of robotic embodiments. Prospective application domains include legged locomotion [30], soft exoskeletons [31], aerial robotics [32], underwater vehicles [33], space robotics [34], and agricultural robotics [35]. In each case, the core safety substrate (Layers 3–4) remains unchanged; only the Hamiltonian, diffusion model, and semantic mapping (Layer 1) are domain-specific.

7.4 Digital Twins and Real-Time Model Updates

A natural extension of the EA-HC framework is to replace the fixed plant model in Layer 4 with a digital twin that is updated in real time from sensor feedback [49, 50]. This would allow the admissibility condition to track changes in the system’s energy landscape (e.g., tissue stiffness changes during a surgical procedure [42]) and adjust the stochastic buffers accordingly. Physics-Informed Neural Networks (PINNs) [36] offer a promising mechanism for online Hamiltonian identification without requiring explicit physical models.

7.5 Semantic Perception and LLM Integration

The current implementation of the semantic mapping $\mathcal{M}$ is rule-based. A natural extension is to leverage large language models (LLMs) to parse nuanced task specifications (e.g., “exercise extreme caution near the aortic valve”) and translate them into graded energy ceilings $E_{\max}$. Recent work on LLM-based task planning for robotics [37] provides a foundation for this integration, though the mapping from linguistic hedges to energy bounds remains an open research problem.

7.6 Swarm and Quantum Extensions

For multi-agent microrobot swarms, the EA-HC admissibility condition extends naturally to a sum of per-agent tank levels, with a global passivity constraint enforced at the swarm coordinator [3, 18]. At the quantum scale (below 10 nm), the classical Hamiltonian H must be replaced by the expectation of the quantum Hamiltonian operator $\hat{H}$, and the Itô calculus must be replaced by its quantum analogue [38]. These extensions are identified as long-horizon research directions.

7.7 Limitations

7.7.1 The Passive Drift Problem

A primary limitation of the current thermodynamic focus is what we term *geometric blindness*. While the EA-HC ensures internal passivity (preventing the robot from becoming mechanically unstable through its own actuator actions), it does not guarantee spatial survival. A robot can be perfectly passive and yet be pushed into a vessel wall by exogenous ambient forces, such as pulsatile blood flow or pressure gradients [24, 27]. In this iteration, we have deliberately excluded exogenous drift terms to preserve the mathematical purity of the Itô–Lévy energy ledger [10], recognizing that spatial confinement is a distinct safety objective from passivity enforcement, and one properly addressed by Control Barrier Functions [17].

7.7.2 Model-Conditional Determinism and the Sensing-Safety Paradox

The safety veto is deterministic in its decision logic but remains conditional on the fidelity of the plant model $H(x)$. If the Hamiltonian or input matrices are misspecified, the energy tank may track a phantom energy level that does not reflect the true physical state [51]. Furthermore, attempting to model every external force introduces a sensing-safety paradox: the deterministic guarantee becomes vulnerable to fallible environmental sensing (e.g., sensor occlusion or calibration drift [52]). The framework trades exhaustive modeling for mathematical rigor by explicitly bounding the scope of the veto rather than attempting to account for all possible perturbations.

7.7.3 Computational Latency

The real-time evaluation of the two stochastic integrals ($\delta_{\text{Itô}}$ and $\delta_{\text{Lévy}}$) at high control frequencies (kHz) introduces non-trivial computational overhead [53]. Any latency in the veto calculation creates a temporal safety gap: the robot is briefly unshielded between the intent generation and the energy admissibility check. Efficient approximations of the Lévy integral (11) (e.g., via moment matching [54] or sparse quadrature [55]) are a necessary direction for practical deployment.

7.7.4 Absence of Numerical Simulation

The current work is a theoretical architecture paper. No numerical simulations or hardware experiments are reported. The buffer expressions (20) and (21) are derived analytically and are

evaluable from known physical constants, but their practical tightness (i.e., whether they are conservative or tight energy bounds) has not been tested. Simulation studies using the symplectic Euler–Maruyama scheme (13) are in progress and will be reported in a subsequent version.

7.7.5 CMDP Policy Convergence

Layer 2 assumes access to a converged CMDP policy π satisfying the energy constraint (7). In practice, RL training may produce policies that violate the constraint during exploration, increasing the frequency of Layer 3 vetoes [6, 41]. The interaction between CMDP training dynamics and the energy tank’s veto rate is an open problem.

7.7.6 Semantic Mapping Fidelity

The mapping $\mathcal{M}$ from symbolic task specifications to energy ceilings (Section 4.1) is currently rule-based and domain-specific. Its correctness is not formally verified: an incorrectly specified $E_{\max}$ (too high or too low) will propagate through the entire stack. Formal methods for verifying the semantic layer against physical constraints [43, 44] remain an open problem and represent a natural direction for future work.

7.7.7 Jump Intensity Calibration

The Lévy intensity ν in (11) must be calibrated from physiological or environmental data [12, 27]. Miscalibration of ν will cause the $\delta_{\text{Lévy}}$ buffer to be either overly conservative (leading to unnecessary vetoes) or insufficiently cautious (allowing energy violations during rare large jumps). Online calibration of ν from streaming sensor data, using e.g. sequential Monte Carlo methods [56], is a necessary engineering challenge.

7.7.8 Tank Depletion Under Sustained Disturbance

If the stochastic disturbances are sustained over a long horizon, the cumulative buffers $\delta_{\text{Itô}} + \delta_{\text{Lévy}}$ may deplete the tank even in the absence of any control action. In this regime, Layer 3 would be forced to veto all actions (safe fallback $u_{\text{safe}} = 0$), effectively immobilizing the robot. A tank replenishment strategy (e.g., via energy harvesting or planned passive phases) is required for sustained operation.

8 Conclusion and Future Work

This paper has proposed the EA-HC framework as a four-layer architecture for resolving the Handshake Gap between symbolic task intent and physical safety in robotics. The key architectural insight is that energy is the universal currency bridging these two levels: a symbolic task specification can be translated into a thermodynamic budget, and passivity can be enforced deterministically through an energy veto derived directly from the Itô–Lévy formula.

The main theoretical contributions are: (1) the extension of the finite-dimensional SPHS formulation to the jump-diffusion regime (Section 2); (2) the extension of the port-Hamiltonian energy tank admissibility condition by two stochastic buffers ($\delta_{\text{Itô}}$ and $\delta_{\text{Lévy}}$); (3) the Compositional Passivity and Stochastic Safety propositions establishing closed-loop energy bounds; and (4) the dimensional homogeneity observation showing that the architecture applies across seven orders of magnitude in robotic operating scale.

Medical microrobotics serves as an extreme-case instantiation, showing that the architecture remains evaluable and physically meaningful in Brownian-dominated, jump-diffusion environments.

Future Work. Future research will focus on three key expansions:

1. **Geometric integration via CBF layer.** We intend to integrate a Control Barrier Function (CBF) layer [17] to provide geometric guardrails that complement the thermodynamic circuit breaker, directly addressing the passive drift problem (Section 7.7.1). The CBF layer would enforce forward invariance of a spatial safe set (e.g., vessel lumen) against exogenous ambient forces, while the EA-HC Layer 3 continues to enforce thermodynamic passivity. The two layers address orthogonal safety objectives and are designed to compose without conflict.
2. **Semantic mapping via large language models.** To refine Layer 1, we propose using LLMs to translate nuanced surgical or operational instructions (e.g., “exercise extreme caution near the aortic valve”) into precise, real-time energy ceilings $E_{\max}$, grounding the semantic layer in natural language understanding [37].
3. **Physics-informed model identification.** To mitigate model-conditional risks (Section 7.7.2), we will explore the use of Physics-Informed Neural Networks (PINNs) [36] to learn and update the system Hamiltonian $H(x)$ during live deployment, enabling the admissibility condition to track the true energy landscape of a time-varying plant.

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